

KRASNYE BAKI, Russian Federation

WMO: 273690

Lat: 57.133N

Lon: 45.167E

Elev: 110

StdP: 100.01

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -24.7	(d) -30.7	(e) 0.2	(f) -27.8	(g) -27.3	(h) 0.3	(i) -24.5	(j) 6.5	(k) -6.6	(l) 6.1	(m) -6.1	(n) 1.0	(o) 320	(p) 0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	29.7	20.5	28.0	19.8	26.2	18.7	21.7	27.9	20.7	26.3	19.7	24.6	2.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.5	24.8	18.7	13.7	23.6	17.8	12.9	22.6	63.8	27.9	60.3	26.3	56.8	24.7	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.2	5.4	4.9	-31.2	32.1	4.0	1.9	-34.1	33.4	-36.4	34.6	-38.6	35.7	-41.5	37.1
			-31.3	23.4	3.9	1.3	-34.2	24.3	-36.5	25.0	-38.7	25.7	-41.6	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-10.1	-9.2	-2.8	5.4	12.7	16.8	19.3	17.0	11.2	4.4	-2.5	-7.5
	DBStd	11.65	7.47	7.13	4.83	4.93	5.11	4.36	3.59	3.99	4.14	4.33	5.60	7.18
	HDD10.0	2878	623	537	398	153	31	3	0	1	33	179	376	543
	HDD18.3	5143	882	770	656	388	186	80	32	75	215	432	626	802
	CDD10.0	917	0	0	0	16	114	206	288	217	70	6	0	0
	CDD18.3	140	0	0	0	0	11	33	62	32	2	0	0	0
	CDH23.3	1205	0	0	0	2	114	303	519	248	19	0	0	0
	CDH26.7	313	0	0	0	0	24	79	141	67	2	0	0	0
(14) Wind	WSAvg	2.4	2.4	2.5	2.7	2.5	2.6	2.4	2.1	2.2	2.3	2.6	2.5	2.5
(15) Precipitation	PrecAvg	654	48	35	37	36	47	80	69	73	59	66	54	52
	PrecMax	912	95	61	91	99	104	174	164	162	123	180	106	98
	PrecMin	487	12	8	4	11	16	21	5	1	15	14	7	14
	PrecStd	86	17	15	20	23	20	37	39	44	29	38	22	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.3	11.5	22.8	29.4	31.7	32.2	31.7	25.6	18.1	10.1	5.7
		MCWB	1.4	1.4	6.1	13.5	18.4	20.9	21.4	22.2	17.3	12.7	8.7	5.3
	2%	DB	1.2	2.0	7.2	19.6	26.4	28.9	29.9	28.4	22.8	14.2	7.2	3.2
		MCWB	0.6	0.8	3.2	11.7	16.7	19.6	21.0	20.4	16.2	10.8	6.1	2.4
	5%	DB	0.6	1.0	5.1	16.3	23.9	26.8	28.2	25.9	20.3	12.1	5.4	1.6
		MCWB	0.0	0.3	1.9	10.0	15.1	18.9	20.2	19.1	15.0	9.9	4.6	0.9
	10%	DB	-0.4	0.2	3.4	13.8	21.5	24.8	26.4	23.9	18.0	10.4	3.8	0.7
		MCWB	-1.0	-0.5	1.1	8.4	14.0	17.6	19.4	18.1	14.0	8.8	3.1	0.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.0	6.6	14.0	20.1	22.1	23.3	23.8	18.5	13.4	8.6	5.4
		MCDB	2.0	2.9	11.2	20.9	27.0	28.4	29.3	31.2	23.4	16.7	9.6	5.7
	2%	WB	0.7	1.1	3.9	12.2	17.9	20.9	21.9	21.3	16.8	11.5	6.5	2.5
		MCDB	1.0	1.6	6.2	18.5	24.5	27.4	28.4	26.5	21.3	13.5	7.1	3.0
	5%	WB	0.1	0.4	2.4	10.6	16.3	19.8	21.0	20.0	15.5	10.1	4.7	1.0
		MCDB	0.6	1.0	4.3	15.6	22.2	25.4	26.5	24.3	19.5	11.8	5.4	1.4
	10%	WB	-1.1	-0.5	1.4	8.8	15.0	18.5	20.0	18.7	14.3	8.9	3.1	0.2
		MCDB	-0.4	0.3	3.0	13.0	20.0	23.3	24.9	22.8	17.5	10.3	3.7	0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	7.1	7.8	9.4	10.3	9.8	9.6	9.1	8.1	5.5	3.9	4.6
		MCDBR	3.9	4.8	9.3	13.3	13.7	12.8	12.5	12.3	12.2	9.0	5.2	3.3
		MCWBR	4.0	4.3	6.2	7.4	6.0	5.4	5.1	5.5	6.5	6.0	4.6	3.3
	5% WB	MCDBR	4.2	4.4	7.8	12.6	12.3	11.5	11.3	11.2	11.1	8.0	4.7	3.1
		MCWBR	4.3	4.0	5.6	7.3	5.8	5.6	5.3	5.7	6.3	5.9	4.4	3.2
(40) Clear-Sky Solar Irradiance	taub		0.282	0.316	0.351	0.402	0.377	0.373	0.404	0.417	0.365	0.348	0.309	0.277
	taud		2.071	2.053	2.037	2.128	2.365	2.407	2.332	2.274	2.401	2.333	2.198	2.045
	Ebn at Noon		558	701	780	803	860	868	827	785	779	678	532	444
	Edh at Noon		58	92	122	132	111	108	114	113	86	71	52	45
(44) All-Sky Solar Radiation	RadAvg		0.50	1.27	2.71	3.96	5.34	5.60	5.55	4.33	2.57	1.15	0.43	0.27
	RadStd		0.08	0.30	0.30	0.41	0.35	0.47	0.51	0.44	0.34	0.20	0.07	0.07

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-185	N/A	N/A
(47) Regional (7 neighbors)		+0.60	N/A	N/A	N/A	N/A	N/A	-150	-196	N/A	N/A

Nomenclature: See separate page